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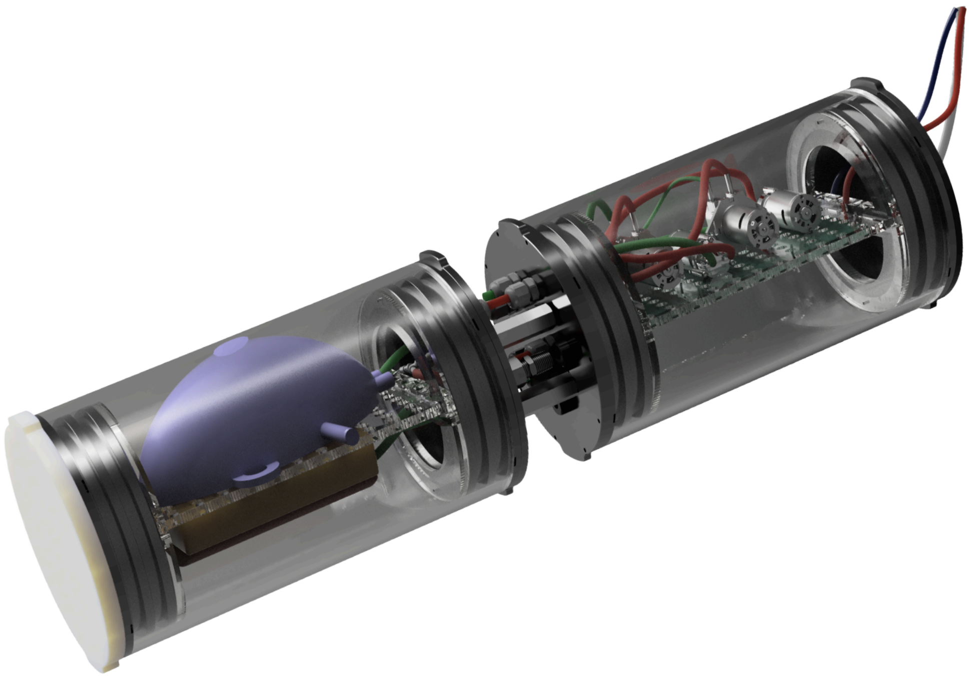

PUMPY, A Test Configuration for an Aluminum-Fueled Buoyancy Engine

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Overview

This project aims to demonstrate the core functionality needed for an aluminum-fueled buoyancy engine. "PUMPY" harnesses the gas generated from a solid aluminum fuel reaction with an aqueous solution of ammonia to change its buoyancy. The developments achieved here reflect an outline that can be iterated upon and modified for implementation into a deployable buoyancy engine which would then be attached to a larger Unmanned Underwater vehicle (UUV).

Background

Existing Solutions and Long-Term Goals

Current methods to energize buoyancy engines in UUVs are hindered by their reliance on lithium-ion batteries. Traditional lithium-ion batteries only support 3 days of AUV operation at 3 knots, failing to meet the demands of long underwater missions that can last weeks or months [1]. While gliders can ride ocean currents at a particular depth for months at a time, changing buoyancy states consumes significant energy, and frequent depth changes will constrain a glider's battery supply [2]. Surface ships are deployed to monitor UUVs, but at $70,000 to operate per day, this greatly raises lifetime cost. [4] Broadly speaking, UUV energy storage systems fall short on operational range. The PUMPY testing configuration lays the groundwork for a buoyancy engine that consumes significantly less battery power, utilizing a chemical reaction and extending range compared to those relying exclusively on batteries.

Basic Operation

Buoyancy engines displace fluids to change their buoyancy, and in turn, that of a greater vehicle. As depicted in Figure 1, a UUV buoyancy engine would initiate a chemical reaction between aluminum and water which produces hydrogen, aluminum byproduct, and a significant amount of heat which is shown in equation 1. The heat is then used to vaporize the ammonia creating a significant amount of gas. The gas-to-water ratio within the engine can be changed by creating or expelling gas, resulting in net positive buoyancy ($Fb > Fg$) for ascent, net negative buoyancy ($Fb < Fg$) for descent, or neutral buoyancy ($Fb = Fg$).

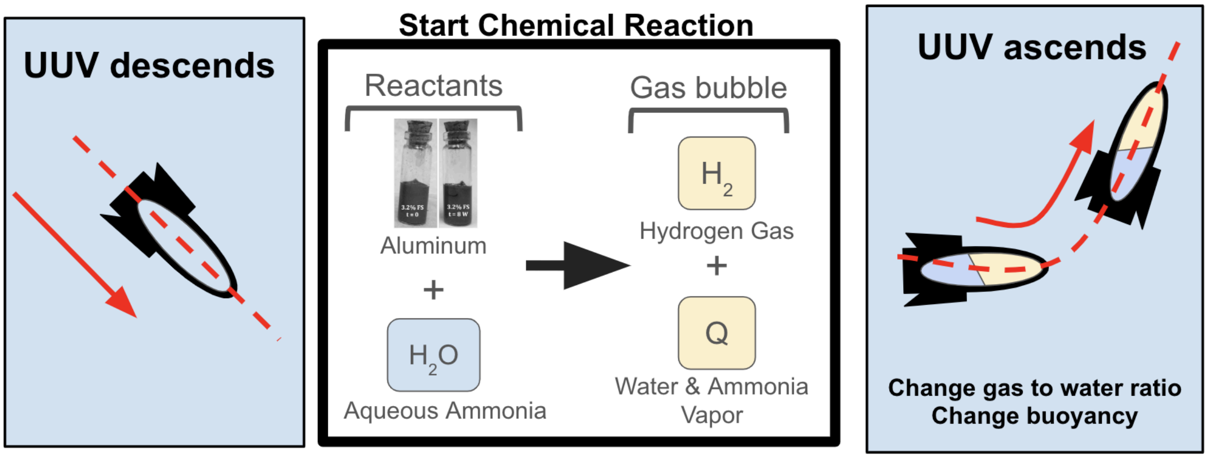


Figure 1: A *diagram of how a UUV would utilize a chemical reaction to increase its gas-to-water ratio. This displaces water volume to change the vehicle's net buoyancy, allowing transition from descent to ascent. The aqueous Ammonia solution is 20% ammonia by mass.*

$$2\,Al_{(S)} + 6\,H_2O \rightarrow 3\,H_{2(g)} + 2\,Al(OH)_{3(aq)} + Q \qquad (1)$$

Project Scope

High Level Objectives

This team set out to fabricate a test configuration that demonstrates the critical aluminum water reaction within a waterproof enclosure. The team then planned to analyze how such a configuration could be scaled up to handle the conditions necessary for true ocean exploration.

System Requirements

Table 1 shows a list of key functional requirements comparing our test configuration to an ocean ready deployment upon a UUV. These specifications were reduced in scope to facilitate rapid production of a proof of concept system.

Table 1: System Requirements for PUMPY

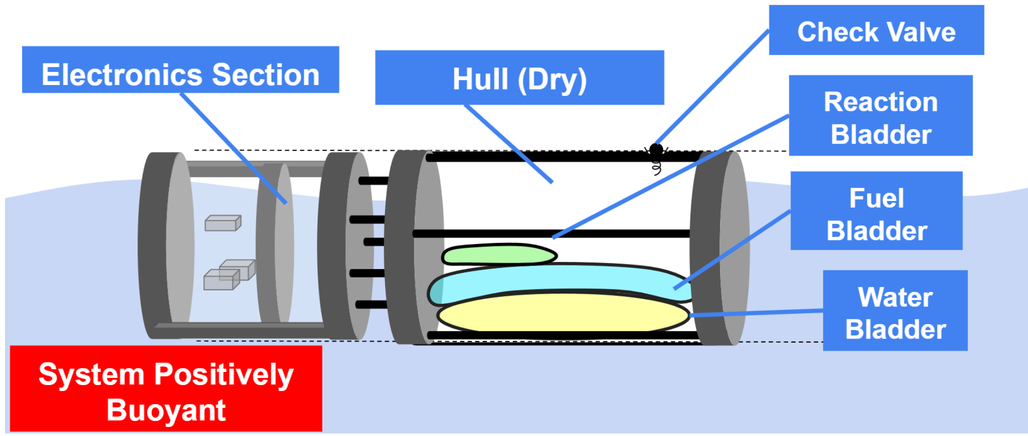
Requirement	Description	Theoretical	Prototype
Operational Depth	Maximum dive depth	**500 m**	**5m**
Operational Pressure	Pressure range of operation	**1 - 50 atm**	**1 - 1.5 atm**
Operational Temperature	Temperature range of operation	**0 - 49° C**	**15 - 49° C**
Vehicle Compatibility	Max System Diameter	**< 0.15 m**	**N/A**
Safety	Maximum H_2 gas concentration from leakage	**< 3%**	**< 3%**
Buoyancy Displacement	Required volume of water displacment from reaction	**≥ 350 mL**	**≥ 350 mL**
Moment Control	Fine control over angle of elevation	**Depends**	**N/A**
System Mass	Maximum system mass	**5.9 kg**	**5.9 kg**

The system requirements were modified so that a full test of the contained, underwater reaction could be set-up, run, and deconstructed within about 3 hours. These changes were also made so that a typical pool could serve as a sufficient testing environment.

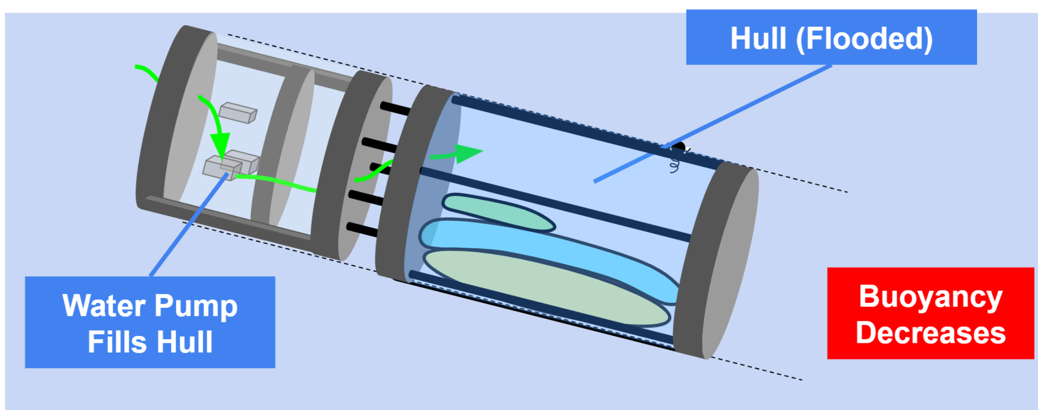
As a follow up to the test configuration, several key changes would be necessary for performance in the real ocean environment and depth. Hardware rated to withstand greater pressure and a higher stored reactant volume would allow sustained dives at depth. A removal of the current tether for onboard power supply and a decrease in design volume, and/or consolidation into a single chamber would be needed for compatibility with typical UUVs. The only change that is not intuitively scalable is the chemical reaction. The reaction in the current test configuration used to inflate the bladder is that of aluminum and water. Compared to water, ammonia has a lower enthalpy of vaporization, will expand with less energy, and condense much more slowly, all of which would be more favorable for repeated reactions especially at depth. In the context of the testing, which aims to verify if the core mechanisms of this engine can function reliably, these benefits are not worth the extra handling caution needed for ammonia. While the water and aluminum reaction will be used for prototyping, we highlight that this ammonia material change will be important for reliable low depth operation.

Figure 2 shows an overview of how the integrated system will work. It outlines a 5-step cycle showing how the buoyancy engine will descend and ascend using a combination of pumps, bladders and the critical aluminum and ammonia chemical reaction.

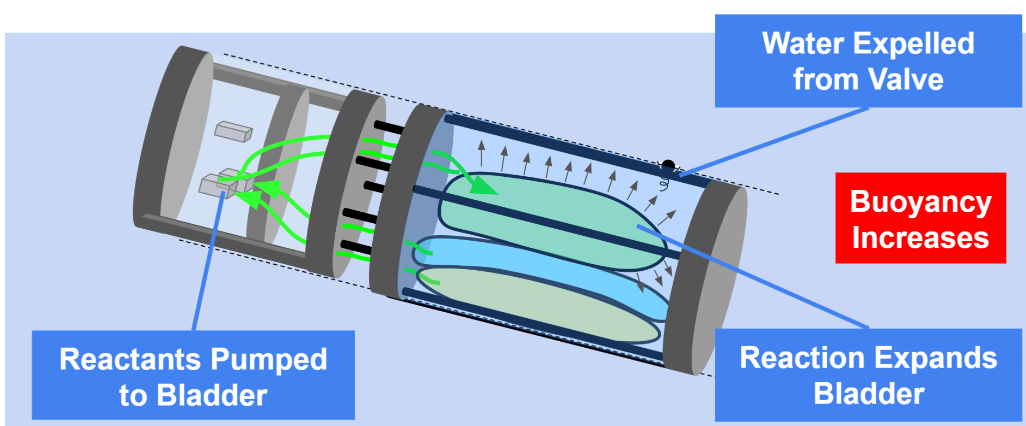
Step 1 System at Rest



Step 2 Descending



Step 3 Ascent (Reaction Start)



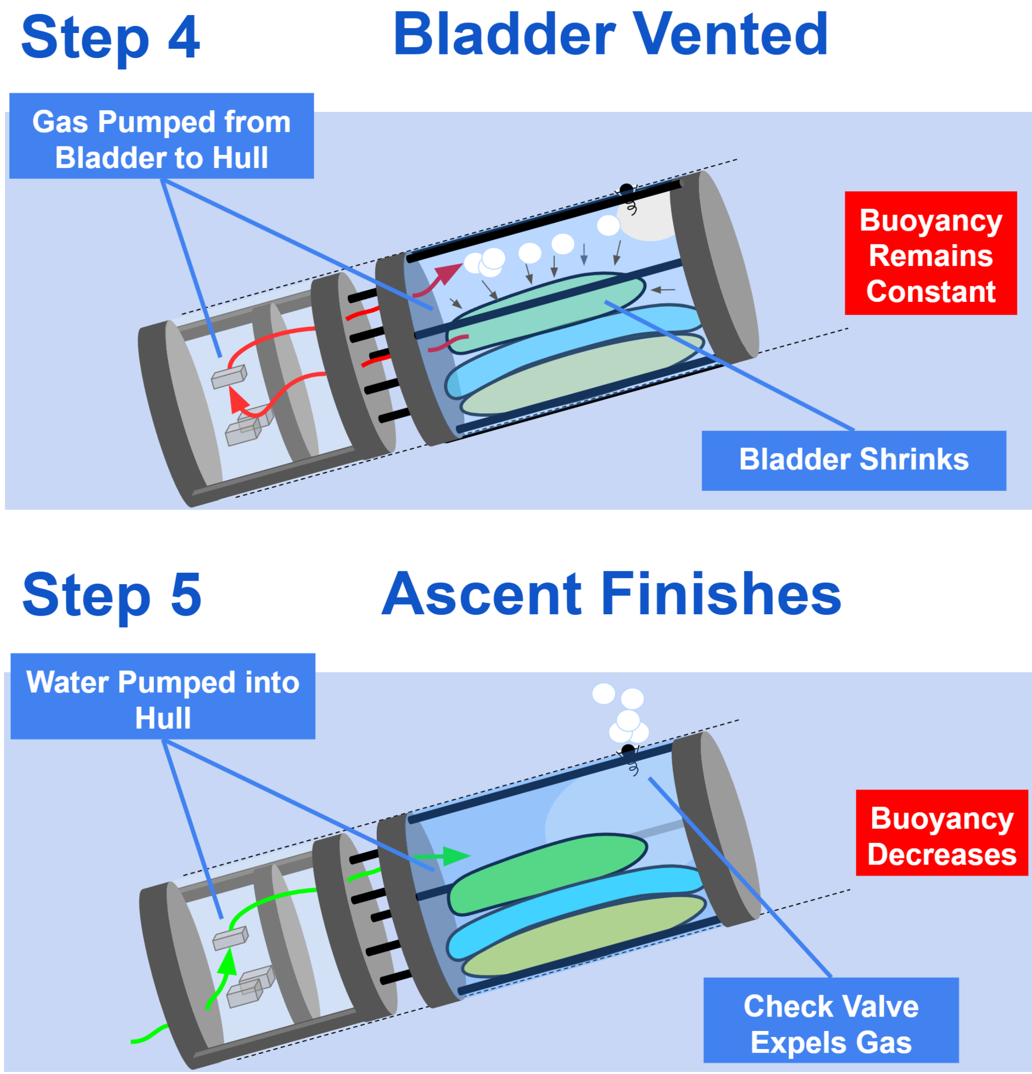


Figure 2: *The dive cycle of the PUMPY prototype*

Approach

To demonstrate controllable buoyancy force via a solid aluminum reaction, the team divided the effort into five separate subsystems: (1) Reaction management, (2) Reaction control, (3) Electronics, (4) Mounting & Enclosure, and (5) Pressure Equalization. Each of these subsystems builds towards the goal of either enabling or directly controlling the aluminum reaction. Figure 3 provides a visual of the hardware planned for this assembly, with two separate chambers, one kept dry and one flooded. One chamber is kept dry to house the electronics, those being four pumps and a PCB. The other is flooded with water, but is still sealed from outside water in order to precisely control flow and pressure within.

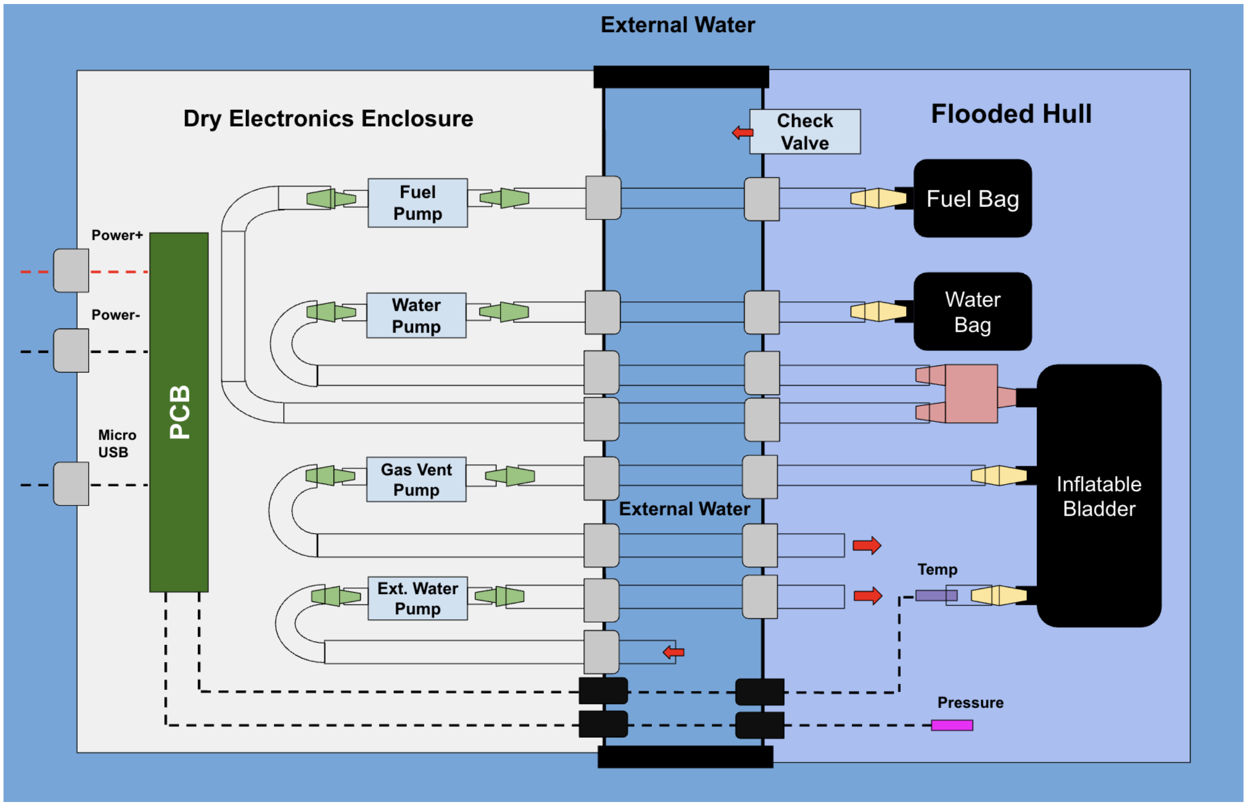


Figure 3: Block diagram of *PUMPY*

System requirements were defined as a whole and then splitted into the defined subsystems to focus on material sourcing, assembly, and testing. Weekly meetings were used to ensure areas of integration were aligned throughout the term. Once individual subsystems were nearing completion, the team met together more frequently to merge the systems into a cohesive testing body. Testing conditions were discussed and planned to ensure meaningful results could be obtained from running the reaction. The questions the team set out to answer were as follows:

- Can we repeatedly demonstrate a change in buoyancy imposed by the gas generated from an aluminum-water reaction?
- What compromises must be made for a testing configuration that can be assembled quickly and iterated upon but will still translate well to an ocean deployed system?
- How will our testing configuration meaningfully inform the design, material selection, and control strategy of a real aluminum-fueled buoyancy engine?

These guiding questions informed the experiment, subsystem work, and testing measures used throughout the semester.

Work Completed

Reaction Management

The reactant management system aimed to reliably store reactants within the flooded hull of the enclosure, deliver and mix precise amounts of reactants ensuring predictable reaction rates, and utilize the gas products from the reaction to displace water volume within the hull and thereby generate positive buoyancy.

The first task, reactant storage, was solved through the flexible storage bags that would allow for a zero pressure gradient between the flooded hull and the stored reactants, allowing the storage compartments to shrink over time as reactants are consumed. Chemical compatibility was assessed and proper polymer materials selected for each bag based on the high concentration ammonia and fuel slurry mixture. For the current design, since water is being used in place of the ammonia solution, the bags are of similar construction. Both storage compartments are fitted with pressure fittings leading into two separate tubing pathways by which they are conveyed into the reaction chamber.

The second task of delivering and fully mixing precise amounts of reactants has been solved via a combined mechanical and controlled effort. The team selected a micro-diaphragm pump to convey the water solution with a control scheme allowing for fine resolution over the amount delivered to the reaction. Conveying of the fuel suspension presents additional challenges given its nature as a suspension and the larger current particle size achieved for the fuel particles. Currently, average fuel particle size is approximately 10 microns compared to the final goal of 5 microns, therefore a higher viscosity oil was used to ensure the particle remains in suspension. To successfully convey this viscous suspension a peristaltic pump was selected to ensure the coarse mixture does not interfere with the pump's operation. The two reactants mix under laminar flow conditions at a y-joint and the reaction initiates. The reaction continues to take place within high temperature tubing downstream of the y joint.

Finally, the third challenge of displacing water volume from the hull was solved by designing a flexible bladder that will readily expand when the reaction produces the gas products. The gas produced via the reaction in the tubing just upstream will expand and fill the bladder over the course of the reaction. The expanding bladder will increase the pressure in the chamber until it reaches the cracking pressure of the check valve that regulates flow between the hull and outside environment. Once this pressure is reached the check valve opens and the water is pushed out of the hull until the pressure within the hull equals the outside pressure. The check valve ensures one way flow to prevent water from filling the hull again once gas volume is lost due to water vapor condensing as the reaction products cool.

The reaction management system provides the mechanical configuration needed to test the efficacy of using the aluminum reaction to produce positive buoyancy. This configuration is designed to de-risk the mechanical execution of a chemical reaction buoyancy engine using the powdered aluminum fuel slurry.

Reaction Control

A manual open loop control system was used for the actuation of four pumps managing water and aluminum paste dispensing, gas venting, and water intake. Pulse width modulation (PWM) was used to allow for varying the rate of material dispensing based on pump needs. Hull pressure and temperature sensors serve as passive monitoring tools to verify expected system behavior, though they do not actively influence control decisions. A Blue Robotics pressure sensor is placed in the internal hull to measure pressure increases due to the chamber filling or the bladder inflating. One thermistor placed inside the bladder checks for temperature increases due to the reaction, while another thermistor inside the hull acts as a temperature setpoint to reduce noise in sensor readings. This approach is straightforward but still ensures precise reactant delivery and bladder deflation.

The system relies on time-based sequencing and predefined actuation commands rather than sensor-driven feedback. External commands provide triggers for each phase of the cycle. These commands consist of ascend, descend, abort ascent, and abort descent. During a test, monitor hull pressure and temperature can be monitored in real time to confirm when a reaction is happening.

The team developed code within the Arduino IDE, chosen for its user-friendly interface and easy deployment. This code sequences the water and aluminum paste dispensing, gas venting, and water intake, with an ESP32 microcontroller delivering the digital outputs. Abort mechanisms are included in the event of hardware malfunction, unexpected sensor readouts, and for general convenience of pausing without severing communications. A graphical user interface was developed, as shown in Figure 4, for convenient control and monitor during testing.

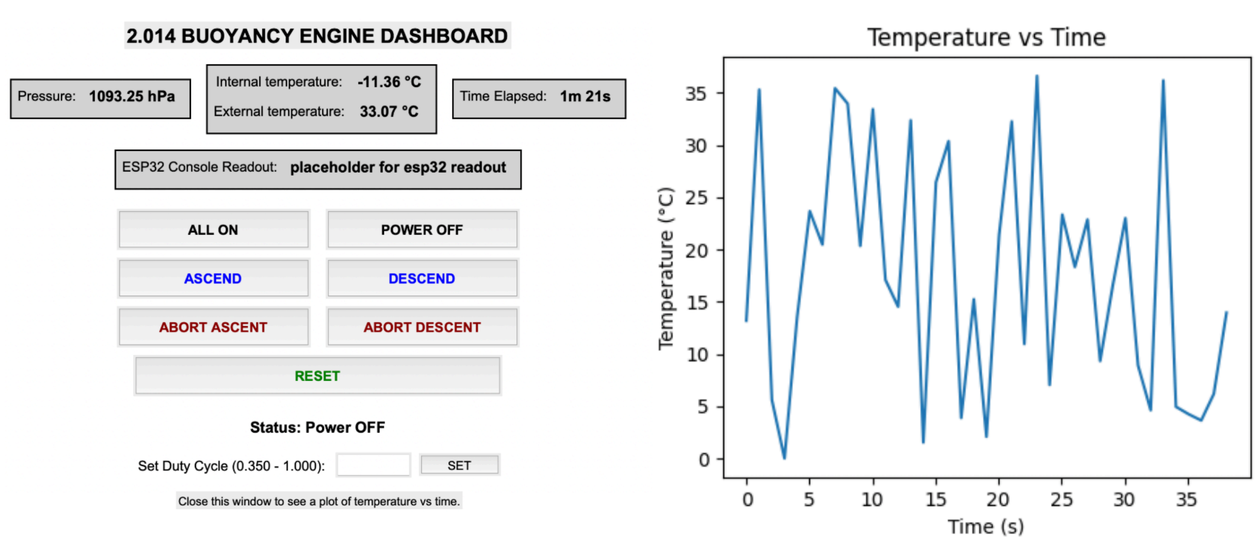


Figure 4: *The buoyancy engine graphical user interface, example data shown.*

Electronics

The electronics system was designed with the requirements of controlling four pumps with sensor readings. As the bidirectional pumps were removed for this prototype, PUMPY will have a simple power MOSFET and gate driver setup. An ESP32 was selected as the microcontroller for its low cost, small size, and robust peripheral set. A standard LM2576 buck converter has also been included to handle any external power sources up to 40V. For testing, a standard rack power supply set to 12V is expected to be sufficient for powering the system hardware. The system can operate solely through the onboard firmware of the ESP32 while standard serial communication protocols are present such as USB, CAN, and SPI for extra function integration like thermal and pressure sensors. The electronics are housed on a single, rectangular PCB, shown in Figure 5, shaped to fit on the middle mounting board within the cylindrical body of the engine. For testing purposes, a tether and a micro usb cable are being used for power and communication, which consists of 3 cables total that are routed into the back side of the electronics hull . This allows for more reliable power delivery and data transfer. This choice eliminates engineering effort to evaluate additional hardware, latency, and dropouts that may come with pursuing wireless communication.

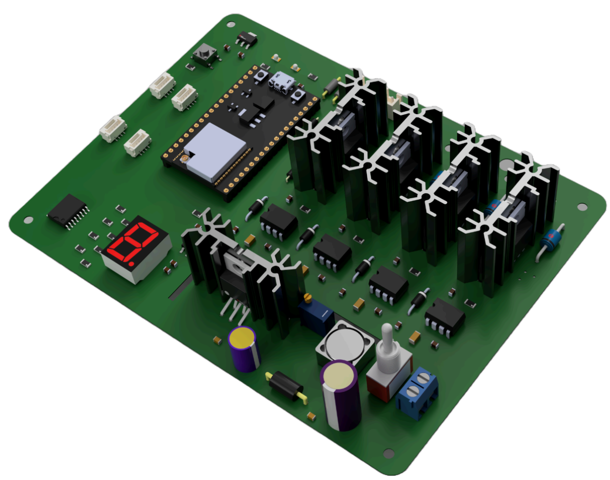


Figure 5: *Visual of the buoyancy engine's PCB*

Enclosure Design

The previous system consisted of a single enclosed tube containing a flooded hull section and an electronics enclosure filled with mineral oil to enhance sealing in deep-sea environments. However, due to the target shift to a proof-of-concept, the team's approach to overall housing evolved significantly. Critical changes are mostly governed by the reduction of operation depth to 5 meters and elimination of the fine buoyancy control and moment balance control features. Design changes concerning enclosure design included reducing the number of pumps and consolidating the system into a single reaction bag instead of two. The previous design required a waterproof interface not only to the outside environment, but within the tube itself.

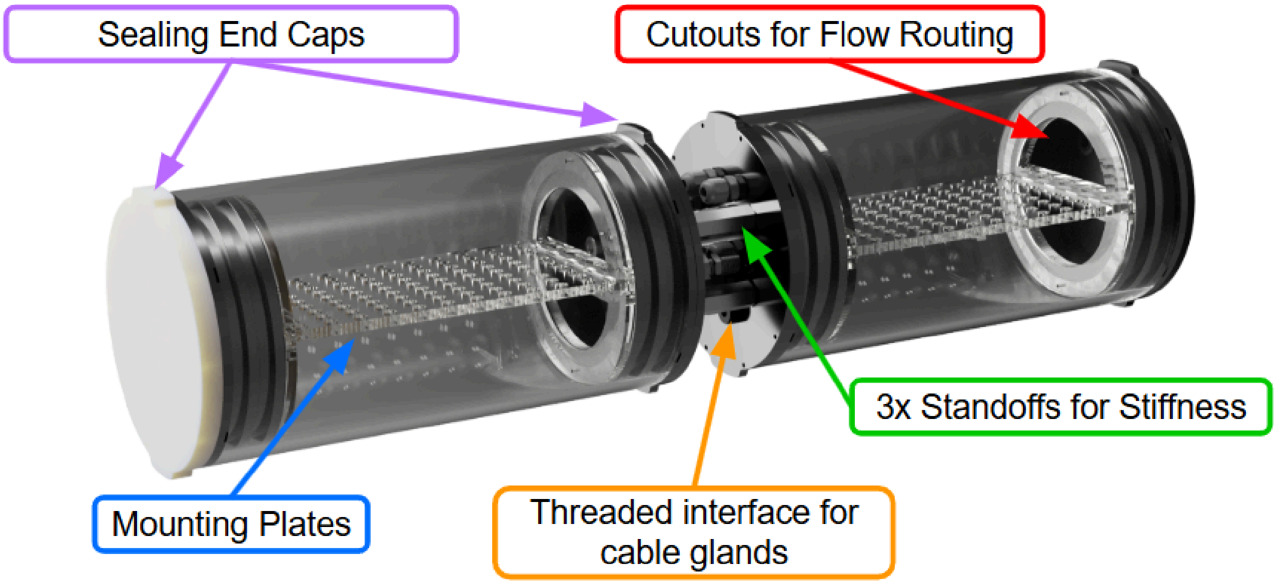


Figure 6: *Enclosure with two independently waterproofed sections*

The internal waterproofing posed a significant manufacturing and installation concern. To mitigate leakage risks, the system was restructured into two independent tubular sections, as shown in Figure 6, both of which were sourced from commercial-off-the-shelf enclosure. Right section is the electronics housing, while the other is the water-flooded hull. Instead of needing to manufacture a custom waterproofing wall between the electronics and the hull, one enclosure now contains the sealed electronics system and the other holds the reaction bladder, fuel, and ammonia storage. These enclosures are connected rigidly via steel coupling nuts and with metal cable glands that seal reactant tubings as they transfer between the electronics enclosure and the water-flooded hull.

The hull section features an acrylic mounting plate where all components are secured. This plate is placed horizontally between two vertical end plates, all held in place via dowel pins which fit into premade enclosure flanges. This modular design significantly improves assembly and disassembly speed for prototyping, ensures water-tight integrity, and decreases manufacturing complexity. The electronics housing has a similar internal structure hull section, except for having additional holes on the other end cap to allow power tether and communication cable to pass through.

Pressure Equalization

Accomplishing the goal of repeatable dives requires the release of gas generated within the reaction bladder outside of the hull to ensure the engine regains negative buoyancy and can dive again. The main challenge of this effort was pumping gas over very large pressure gradients if operating near the max depth of 500 meters, as is the case for the full scope design. This challenge was addressed by splitting the gas venting process into two discrete steps: venting of gases from the reaction chamber into the flooded hull and addition of outside water into the hull to force the gases through the check valve and out of the enclosure. This approach removes the concerns associated with the large pressure gradient. The first step has no pressure gradient as the pressure within the hull does not change while the gases are vented into the enclosure, only the location of the gases themselves is changed from within the reaction bladder to outside of the bladder within the water in the hull. The next step takes advantage of the pressure gradient between the outside and inside of the hull. A separate pump takes in water from outside the hull via a tubing port and then pumps it into the flooded hull. This pump has a pressure head greater than the cracking pressure of the check valve within the flooded hull, increasing the pressure within the hull enough to force the trapped gasses out of the hull. Through this approach, the gas can be expelled without attempting to pump against a large pressure differential and instead only a small pressure head is needed to overcome the check valve.

Assembly and Testing

Enclosure underwater tests were conducted first to verify waterproofing. The setup included 3D-printed caps, acrylic middle plates, and cable glands that provided connectivity between internal components and external control interfaces. For this initial test, all cable glands were sealed with grease, the enclosure's O-rings were lubricated, and Teflon tape was applied to all threaded connections. The preliminary water test, conducted with the enclosure alone, was successful. The team then proceeded to a full system test—excluding fuel—by connecting the pumps and bladders within the enclosure and circulating water through the system.

Testing Apparatus

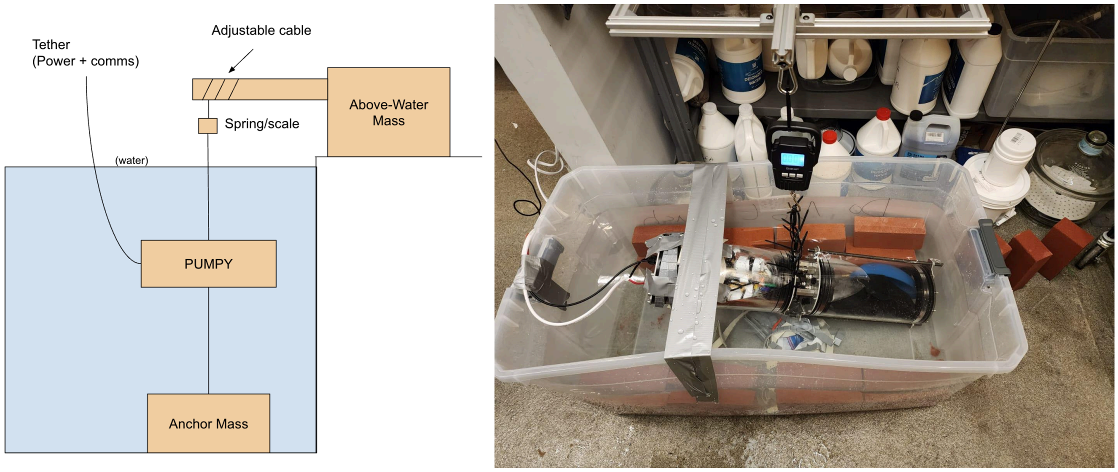


Figure 7: *Testing apparatus schematic and real test*

A testing apparatus was created to allow indoor testing in a lab environment, shown in Figure 7. A hanging scale is suspended from an aluminum frame, to which the testing configuration is attached. This is then weighed down with an anchor mass to secure the setup. Adjustable zip-ties are used to then pull the system into tension. The proposed experimental configuration is readily adaptable to pool- or dock-side facilities that provide a minimum water depth of 5 m, thereby replicating the targeted proof-of-concept operating conditions.

Results

The results of chemical volume requirements as well as change in buoyancy force have been calculated. The chemical volume requirements were derived from the moles of gaseous H_2O required to fill the bladder (350 mL of volume) at 111℃ and 0.5 atm.

- H20 vaporized replaces the ammonia in the ammonia solution which will lead to 1 mol of H20 being vaporized for every 4 mol reacted.
- 111℃ is the boiling point of water at 0.5 atm (5 meter depth)
- This is a minimum calculation to ensure there is a measurable amount of buoyancy change as well as was used to ensure heat dissipation would not affect performance.
- We expect the steam to expand past 350 mL as well as H_2 to contribute to volume increase however this should not be an issue since our 400 ml bladder is able to stretch up to 2x its size.

Using the stoichiometric equation above as well as known densities and molar masses, the amount of fuel and water needed for the reaction is 0.63 mL and 1.5 mL respectively. This reaction will be able to heat the system to 111℃ with an excess of 8.19 kJ ignoring heat transfer. Through a simplification of the bladder geometry the maximum heat transfer rate was found to be 0.127 kJ/s, meaning as long as the reaction happens in less than 64s we will be successful. (Although the exact rate of reaction is not known we will empirically check to see if it is less than 64 s) All assumptions and calculations can be found here in the thermal calculations and chemical calculations pages. Therefore, the amount of buoyancy generated should be greater than 3.5 N of force which will be a measurable change given our experimental setup.

Due to the danger inherent in dealing with the highly reactive and energetic aluminum fuel, a proper substitute slurry was created using similarly sized 316L stainless steel powder suspended in silicon bath fluids. This suspension oil is the same as previously used in the creation of the aluminum fuel slurry, giving this substitute suspension similar properties as the final fuel. The aim of this effort was to accurately determine the kinematic viscosity of the fuel slurry to properly select the pumps that will be used to convey it. These testing results suggest a kinematic viscosity between 150 to 175 cP, which correlates to a required pressure head of approximately 4 KPa based on the selected tubing internal diameter of 3/16 inches, a max tube length of 1 meter, and Hagen-Poiseuille's equation for laminar viscous fluid flow. This was increased by 50% as a safety factor to arrive at a required pump head of 5.8 KPa. Most commercially available peristaltic pumps have pressure heads far higher, therefore low cost reactant pumps were selected given they still achieved a pressure head of 21 KPa and consistent dosing rates .

The testing configuration demonstrates a control system that can adequately dispense reactants for the aluminum-water reaction. This system is capable of repeated cycles trapping gas to increase buoyancy, venting the produced gas into the hull, and then forcing water into the hull to dispel the gas into the environment decreasing buoyancy. The observed repeatability of this function is an important step forward, given the target application of long duration missions with infrequent replenishment. It has been shown that the fundamental sequence of controlled events lends itself well to a simple control system and does not require any particularly advanced feedback control to perform its core functions. While active control would certainly benefit a deployment ready buoyancy engine, demonstrating that it is not essential suggests that the reaction and system as a whole are relatively stable and predictable.

Consistent fuel delivery for a given ascent event is specifically demonstrated. The peristaltic pumps used in the configuration are capable of repeatedly dosing approximately 0.6 mL of aluminum fuel slurry to enable the reaction on command. It was observed that the peristaltic pumps behaved as one-way flow routing, as expected, and as a result, no backflow issues were encountered. The Y-shaped fitting and simultaneous flow of both aluminum paste and water ensured our reaction occurred upstream of the bladder with consistent flow of gas products into the reaction bladder. These results are meaningful indications that this reaction, while volatile, can be harnessed and directed in the confines of a real buoyancy engine.

Relevant hardware changes were made throughout the term to arrive at a coherent, waterproof, and pressure managed system. Upon first trials in water, the system failed to remain completely waterproof. Leakage was spotted at the cable glands used to pass the system's tubing from one enclosure to another. Several changes were made that eventually solved this problem: teflon tape around the threads of the glands, swapping soft tubing for hard nylon plastic tubing, and metal cable glands instead of plastic. With these changes made, the system remained waterproof through all following testing. Another realization was managing pressure buildup within the flooded hull. It was expected to be an immediate outflow of water through the system's check valve, but instead, the chamber's end caps were unseated and pushed off of their tubes. A critical change was retaining both end caps, in this case with external threaded rods, to prevent this behavior. An important target through all of the changes was serviceability. The system avoided more permanent work-arounds, such as epoxy, for sealing as this would not translate well for a buoyancy engine that needs its fuel bags swapped out on occasion. Instead, it is shown that sufficient hardware improvements can lead to a system that is both fully waterproof and easily, repeatedly taken apart.

As depth increases, waterproofing becomes considerably more challenging due to a greater external pressure difference. Several takeaways were found that will still apply well even at depth. First, an understanding of and adherence to sufficiently depth-rated connectors is a solid starting point. An array of commercially available connectors are available, rated to 500 m depth, and even beyond. It was then noted that metal fittings not only have less compliance than plastic ones, but also can handle more fastening torque, all together making them a much preferable general purpose upgrade. It was also found that rigid plastic or metal tubing should be treated as a necessity for any sections exposed to open water. Connectors and depth ratings on their own do not guarantee success, but a proper application of how these components engage with their surroundings is critical to keeping the system waterproof.

A custom silicone bladder was developed for the testing, and several key takeaways were observed as creating and testing this component. The first note was on the design metrics which are critical in handling repeated inflation and deflation, high temperature reaction, and surrounding environment pressure change, while still engaging properly with its connectors and staying air-tight. The geometry and hardness of the material played an important role in keeping the bladder intact. The interfaces for connectors are a major problem area for tears which can completely compromise a bladder. Both in initial installation and after repeated testing, the plugs through which barbed fittings went through were slightly split apart. The bladder was 3D printed, and while this fabrication technique lent itself very well to a custom geometry, attention should be given to the layer lines and orientation of the print. These tears consistently occurred in parallel with the layer lines, which were also in parallel with the barbed fittings. If the print orientation could be changed, going against barb insertion, or at an angle, this would likely reduce the chance of such damage occurring again.

Another useful takeaway from the bladder development which proved successful was having an asymmetric geometry for uneven inflation. While a uniform, spherical bladder may seem intuitive, the oval shaped, flat base geometry ended up providing key advantages. By positioning the connectors towards the base and narrower section of the oval, less inflation occurs at the connectors, and the seals are maintained. If this was a sphere, or if the connectors were at the wider oval region, they would be much more liable to unseating as more expansion occurs at their interface. These lessons learned in bladder design will apply generally to multiple fabrication techniques and will minimize the chance of leakages for a buoyancy engine ready bladder.

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